Light

GCSE Computer Science — Networks Basics (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Networks Basics

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does LAN stand for?

- A. Local Area Network
- B. Large Area Network
- C. Long Access Network
- D. Linked Application Node

Q2 (1 mark)

Which type of network spans a large geographical area, such as across cities or countries?

- A. WAN

- B. LAN
- C. PAN
- D. MAN

Q3 (1 mark)

What is the main function of a router in a computer network?

- A. To convert digital signals to analogue signals
- B. To store files and programs for network users
- C. To direct data packets between different networks
- D. To provide electrical power to network devices

Q4 (1 mark)

What does a network switch do?

- A. Connects devices within the same network and directs data to specific devices
- B. Connects a local network to the Internet
- C. Converts wireless signals to wired connections
- D. Blocks unauthorised access to the network

Q5 (1 mark)

What is a NIC (Network Interface Controller)?

- ○ **A.** Software that manages network connections
- ○ **B.** A device that boosts wireless signals
- ○ **C.** Hardware that allows a device to connect to a network
- ○ **D.** A security system that monitors network traffic

Q6 (1 mark)

What is the purpose of a Wireless Access Point (WAP)?

- ○ **A.** To connect wireless devices to a wired network
- ○ **B.** To store data for wireless backup
- ○ **C.** To increase the processing speed of wireless devices
- ○ **D.** To encrypt all network traffic

Q7 (1 mark)

What is the main purpose of a DNS (Domain Name Server)?

- ○ **A.** To convert domain names (URLs) into IP addresses
- ○ **B.** To store website files and content
- ○ **C.** To encrypt data sent across the Internet
- ○ **D.** To block malicious websites

Q8 (1 mark)

In a client-server network, what is the role of the server?

- ○ **A.** To request services and resources from other computers
- ○ **B.** To manage user passwords only
- ○ **C.** To connect the network to the Internet
- ○ **D.** To provide services and resources to other computers

Q9 (1 mark)

What is a characteristic of a peer-to-peer (P2P) network?

- ○ **A.** All resources are stored on a central server
- ○ **B.** A dedicated IT administrator is always required
- ○ **C.** Only one computer can access resources at a time
- ○ **D.** Each computer can act as both a client and a server

Q10 (1 mark)

What does 'the Cloud' refer to in computing?

- ○ **A.** Weather monitoring software
- ○ **B.** The wireless signal between your device and a router
- ○ **C.** A local backup drive connected to your computer

○ **D.** Remote servers accessed over the Internet to store data or run applications

Q11 (1 mark)

In a star network topology, how are devices connected?

○ **A.** Each device is connected directly to every other device

○ **B.** All devices are connected in a circle

○ **C.** All devices are connected to a central hub or switch

○ **D.** Devices are connected in a straight line

Q12 (1 mark)

What does bandwidth measure in a computer network?

○ **A.** The physical width of network cables

○ **B.** The maximum amount of data that can be transmitted per second

○ **C.** The number of devices connected to the network

○ **D.** The strength of the wireless signal

Q13 (2 marks)

A school has computers in different classrooms connected through cables within the building, owned and maintained by the school's IT department.

Which type of network is this, and why?

- ○ **A.** WAN, because it connects multiple classrooms
- ○ **B.** LAN, because it covers a small geographical area and is privately owned
- ○ **C.** WAN, because it uses cables for connections
- ○ **D.** LAN, because it requires the Internet to function

Q14 (2 marks)

A company notices their network has become slower since hiring 50 new employees who all use the network.
Which factor most likely explains the reduced network performance?

- ○ **A.** The network cables have physically degraded
- ○ **B.** The server has been upgraded to a faster model
- ○ **C.** More devices are now sharing the same bandwidth
- ○ **D.** The network topology has automatically changed

Q15 (2 marks)

A business is choosing between a client-server network and a peer-to-peer network for their 200 employees.
Which statement correctly identifies an advantage of choosing a client-server network?

- ○ **A.** Client-server networks are cheaper to set up because no server is needed
- ○ **B.** Client-server networks allow each computer to operate independently without any shared resources

- ○ **C.** Client-server networks work without any network administrator
- ○ **D.** Client-server networks provide centralised backup and security management

Q16 (2 marks)

In a computer lab using a star network topology, one computer's network cable is damaged.
What is the impact on the rest of the network?

- ○ **A.** All computers in the network lose connectivity
- ○ **B.** Only the computer with the damaged cable loses connectivity
- ○ **C.** Half of the network loses connectivity
- ○ **D.** The network automatically repairs the cable

Q17 (2 marks)

A hospital wants a network where data can still be transmitted even if one connection fails, because patient care depends on constant network availability.
Which network topology would best meet this requirement?

- ○ **A.** Star topology
- ○ **B.** Mesh topology
- ○ **C.** Bus topology
- ○ **D.** Ring topology

Q18 (2 marks)

A user types 'www.example.com' into their web browser.
What happens first in the process of accessing this website?

- ○ **A.** The web server sends the website files to the browser
- ○ **B.** A DNS server converts the domain name to an IP address
- ○ **C.** The browser downloads and displays the web page
- ○ **D.** The user's computer connects directly to www.example.com

Q19 (2 marks)

A company moves their data storage from local servers to cloud storage services.
Which is a valid advantage of this change?

- ○ **A.** Data can only be accessed from the office, improving security
- ○ **B.** The company no longer needs an Internet connection
- ○ **C.** Employees can access data from anywhere with an Internet connection
- ○ **D.** All data is automatically printed as a backup

Q20 (2 marks)

When you visit a website, your browser is described as a 'client' and the computer hosting the website is called a 'server'.
Which statement best describes the relationship between client and server?

- ○ **A.** The client stores all website data and the server displays it

- ○ **B.** The server requests resources and the client provides them
- ○ **C.** The client and server have identical roles in the communication
- ○ **D.** The client requests resources and the server provides them

Q21 (2 marks)

A company is deciding between using Ethernet cables or Wi-Fi for their new office network.
Which statement correctly compares these two transmission media?

- ○ **A.** Wi-Fi typically provides faster and more reliable speeds than Ethernet
- ○ **B.** Wi-Fi is more secure because signals cannot be intercepted
- ○ **C.** Ethernet cables are wireless and Wi-Fi requires physical connections
- ○ **D.** Ethernet cables provide more reliable connections but reduce mobility

Q22 (2 marks)

Which statement best describes what the Internet is?

- ○ **A.** A single large computer that stores all websites
- ○ **B.** The same thing as the World Wide Web
- ○ **C.** A private network owned by one company
- ○ **D.** A worldwide collection of interconnected computer networks

Q23 (2 marks)

A new business wants their website to be accessible 24/7 to customers around the world.
What service do they need?

- ○ **A.** Cloud computing for processing
- ○ **B.** Web hosting to store and serve their website
- ○ **C.** A DNS server to create their domain name
- ○ **D.** A firewall to protect their office computers

Q24 (3 marks)

A school is designing a new network and must choose between star and mesh topology.
The school has limited budget but wants easy troubleshooting when problems occur.
Which topology should they choose, and why?

- ○ **A.** Mesh, because it is cheaper to install and easier to troubleshoot
- ○ **B.** Star, because it requires less cabling and individual device failures are easy to isolate
- ○ **C.** Mesh, because it uses a central hub making faults easy to find
- ○ **D.** Star, because it provides multiple redundant paths for data

Q25 (3 marks)

A small startup with 5 employees wants to share files between their computers. They have no IT staff and a very limited budget.
A large corporation with 5000 employees needs to share files with strict access controls and regular backups.
Which network model suits each organisation?

- ○ **A.** Both should use client-server for better security
- ○ **B.** Both should use peer-to-peer for simplicity
- ○ **C.** Startup: peer-to-peer; Corporation: client-server
- ○ **D.** Startup: client-server; Corporation: peer-to-peer

Q26 (3 marks)

A graphic design company is considering moving from local storage to cloud storage for their large project files.
Which evaluation correctly identifies both an advantage and a disadvantage of this change?

- ○ **A.** Advantage: No Internet needed; Disadvantage: Files can only be accessed from the office
- ○ **B.** Advantage: Files accessible from any location; Disadvantage: Requires reliable Internet connection
- ○ **C.** Advantage: Eliminates all security risks; Disadvantage: Storage space is always unlimited
- ○ **D.** Advantage: Files are automatically deleted after use; Disadvantage: No version history available

Q27 (3 marks)

A new office needs to set up their network. They need to:

- Connect 20 computers to each other within the office
- Allow laptops to connect without cables
- Connect the office network to the Internet

Which combination of hardware is needed?

- ○ **A.** Switch only

- ○ **B.** Router and switch only
- ○ **C.** Switch, wireless access point, and router
- ○ **D.** Wireless access point only

Q28 (3 marks)

Aisha types 'www.newssite.com' into her browser. The DNS server returns the IP address 185.199.108.153.
Her browser then displays the news website.
Which sequence correctly describes what happened?

- ○ **A.** Browser → DNS lookup (gets 185.199.108.153) → Request sent to IP → Web server responds with page
- ○ **B.** Browser → Web server responds → DNS lookup → Page displayed
- ○ **C.** DNS automatically sends the website → Browser receives page → IP address assigned
- ○ **D.** Browser → IP address created by browser → DNS not needed → Page displayed

Q29 (3 marks)

A company's network has become very slow. Investigation reveals:

- 100 new IoT devices were recently connected
- The Internet connection is 100 Mbps
- Video conferencing is now used extensively

Which factor is most likely the primary cause of the slowdown?

- ○ **A.** The network cables have become too long
- ○ **B.** The IoT devices have infected the network with viruses

○ **C.** The combined bandwidth demands now exceed the available 100 Mbps connection

○ **D.** The computers need their RAM upgraded

Q30 (3 marks)

Every Network Interface Controller (NIC) has a unique MAC address assigned during manufacturing.
A network administrator needs to restrict network access to only company-owned devices.
How can MAC addresses help achieve this?

○ **A.** MAC addresses can be changed daily to improve security

○ **B.** The switch can be configured to only allow connections from specific MAC addresses

○ **C.** MAC addresses automatically block all external network traffic

○ **D.** MAC addresses encrypt all data transmitted by the NIC

Total: **55** marks • Time: **30 min**

↑ Back to top

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